Supplementary Material for "M^{3}UCD: A Multi-task Multimodal Metaphor Understanding Challenge Dataset for LLMs"

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Over-detection Bias Analysis

Performance-normalized Hallucination Index

Besides the overall detection quality, we further examine whether a model systematically over- or under-predicts the target categories. Specifically, we measure the over-detection bias of a model with the performance-normalized hallucination index $\mathcal{H} = \frac{|Prec-Rec|}{Acc+F1}$, which normalizes the precision-recall gap by the overall detection quality. Here, *Acc* and *F1* are taken from Table 4 of the main paper, and *Prec* and *Rec* denote the corresponding *macro-Prec.* and *macro-Rec.* reported in Table 1 of the Supplementary Material. A lower $\mathcal{H}$ indicates that the model's predictions are better balanced between precision and recall relative to its detection quality, whereas a higher $\mathcal{H}$ indicates a stronger tendency to over-detect (or miss) the target categories. The results are reported in Table 1.

Results Analysis

(I) Zero-shot models exhibit a pronounced over-detection bias. Without task-specific training, LLMs and MLLMs obtain an average $\mathcal{H}$ of 7.53 and 6.54 ($\times 10^{-2}$), respectively, which is substantially higher than that of PTMs (3.38). The bias is concentrated on the metaphor reasoning tasks (8.66 on average for zero-shot LLMs and MLLMs) and the semantic orientation tasks (7.64), and is much milder on the pragmatic figurativeness tasks (3.40). Extreme cases include LLaMA3.1-8B-Instruct on IR (50.66), InternVL2-1B on TDD (46.56), and GPT-4.1 on IR (46.20), where the precision-recall gap reaches nearly half of the overall detection quality. Moreover, a larger model scale or stronger general capability does not guarantee balanced predictions: GPT-4.1 (14.33) and GPT-4o (8.08) are among the most biased zero-shot models, whereas Llama-3.2-11B-Vision-Instruct (2.59) and Qwen2-VL-7B-Instruct (2.80) are the most balanced.

(II) Fine-tuning effectively mitigates the over-detection bias, and MCLF further reduces it. LoRA fine-tuning decreases the average $\mathcal{H}$ of LLMs from 7.53 to 1.87 and that of MLLMs from 6.54 to 1.49, indicating that task-specific supervision calibrates the decision boundaries of the models. Built on the same backbones, MCLF further lowers the average $\mathcal{H}$ of MLLMs from 1.49 to 1.15 (a relative reduction of 23.0%), with the largest gains on the metaphor reasoning tasks, i.e., MC (-1.54), SDD (-1.40), and TDD (-0.53). This suggests that the joint optimization in MCLF not only improves detection performance but also yields more balanced predictions. The only exceptions are IR (+0.71) and HumorD (+0.14), where the $\mathcal{H}$ values of both LoRA and MCLF remain at a low level.

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Type	*Model*	*Metaphor-rich Semantic Orientation Tasks*			*Metaphor-rich Pragmatic Figurativeness Tasks*			*Partial Metaphor Reasoning Tasks*				*Avg.*
		EA	*OD*	*IR*	*HumorD*	*SD*	*HyperD*	*MD*	*SDD*	*TDD*	*MC*	
	Random	1.97	1.53	1.69	0.03	0.03	0.02	0.03	0.32	0.33	1.10	0.70
PTMs	Bi-LSTM	8.30	25.45	6.28	2.08	4.33	0.64	3.04	4.83	0.79	3.66	5.94
	BERT	0.60	8.68	4.96	0.04	0.68	1.07	0.16	3.86	1.21	1.84	2.31
	ResNet50	1.60	2.15	0.63	0.41	2.90	0.72	0.37	3.39	0.64	0.27	1.31
	ViT	11.38	10.11	6.06	2.89	0.13	0.51	0.12	7.30	0.42	5.80	4.47
	MAE	1.35	12.75	3.53	3.66	0.90	0.12	2.30	1.92	1.78	12.16	4.05
	DeiT	4.43	5.17	3.46	0.24	7.20	0.64	1.12	2.40	0.93	7.11	3.27
	SwinT	9.43	7.61	4.55	0.34	1.97	0.34	0.99	5.45	5.27	6.94	4.29
	CLIP	0.98	4.11	0.51	0.01	0.31	0.19	0.31	2.28	0.14	1.06	0.99
	FLAVA	0.05	3.00	2.66	0.17	4.40	1.32	0.74	5.47	4.14	15.97	3.79
LLMs	ChatGLM3-6B-Base	22.88	2.14	14.20	2.15	2.00	0.24	2.11	9.92	9.78	0.09	6.55
	LLaMA3-8B-Instruct	2.37	1.02	1.93	1.58	2.97	0.00	1.19	7.64	17.20	5.07	4.10
	InternLM2.5-7B-Chat	12.40	6.27	7.79	0.21	2.26	0.13	0.26	1.50	3.29	43.10	7.72
	LLaMA3.1-8B-Instruct	13.47	0.27	50.66	2.80	0.71	0.20	0.41	4.13	3.61	5.68	8.19
	MiniCPM3-4B-Chat	6.20	0.37	11.05	1.00	2.30	2.97	0.12	44.89	9.11	18.08	9.61
	GPT-3.5-turbo	0.45	2.90	15.82	3.84	1.21	0.22	0.19	41.58	3.49	20.71	9.04
MLLMs	InternVL2-1B	0.98	3.39	27.34	12.64	7.37	43.63	5.61	7.89	46.56	39.33	19.47
	InternVL2-2B	9.57	5.01	1.71	1.27	11.47	0.35	0.23	0.95	21.60	5.88	5.80
	InternVL2-4B	1.53	4.89	3.02	0.17	6.40	1.74	18.04	8.12	16.29	4.69	6.49
	LLaVA1.5-7B-Chat	4.54	2.06	7.18	0.58	0.50	0.00	1.19	3.23	13.94	2.66	3.59
	Qwen2-VL-7B-Instruct	2.62	2.19	17.07	0.99	0.26	2.08	0.21	1.42	0.72	0.45	2.80
	InternVL2-8B	13.90	11.49	3.02	0.34	5.12	0.21	3.79	1.38	6.45	2.52	4.82
	Llama-3.2-11B-Vision-Instruct	0.76	1.39	8.54	1.19	3.60	0.28	8.17	0.03	0.04	1.93	2.59
	InternVL2-26B	0.19	20.27	3.38	1.51	5.09	0.23	13.93	12.01	8.15	0.27	6.50
	InternVL2-40B	0.23	5.81	8.79	0.34	7.93	0.17	14.20	4.62	5.34	0.43	4.78
	Qwen2-VL-72B-Instruct	0.22	10.35	1.91	0.30	11.36	1.87	9.33	0.68	0.56	12.12	4.87
	InternVL2-Llama3-76B	4.21	0.43	4.21	4.59	4.84	8.47	1.85	7.34	5.55	0.93	4.24
	GLM-4V-Plus	26.00	6.51	4.70	0.17	4.93	0.38	6.43	23.85	4.96	10.92	8.89
	GPT-4o	5.83	8.02	2.55	0.51	2.45	9.12	8.52	22.03	14.88	6.92	8.08
	Gemini-1.5 Pro	1.97	13.84	1.40	1.31	3.74	6.22	13.94	0.09	3.97	18.06	6.46
	Claude-3.5-Sonnet	0.17	7.73	8.71	0.79	4.52	0.08	16.60	1.52	0.74	5.05	4.59
	o3-mini	0.70	9.30	8.58	5.26	13.63	0.27	1.34	0.38	4.39	3.38	4.72
	Claude-3-7-Sonnet-Thinking	2.90	13.00	4.14	2.27	4.29	0.64	13.70	0.54	0.96	1.53	4.40
	GPT-4.1	3.65	15.16	46.20	0.17	0.08	13.00	18.48	13.72	15.53	17.28	14.33
	Gemini-2.5-flash-preview	3.32	5.94	8.05	0.51	6.12	1.13	9.81	4.52	7.52	21.09	6.80
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	0.54	4.24	2.60	0.49	0.60	0.62	0.22	3.36	2.47	3.29	1.84
	LLaMA3-8B-Instruct (LoRA)	0.20	3.39	1.29	0.11	0.79	0.57	0.18	2.29	2.49	3.11	1.44
	InternLM2.5-7B-Chat (LoRA)	0.03	6.29	2.63	0.55	1.01	0.23	0.12	1.64	1.64	3.09	1.72
	LLaMA3.1-8B-Instruct (LoRA)	0.42	3.60	2.70	0.40	1.24	0.39	0.17	7.05	1.07	2.25	1.93
	MiniCPM3-4B-Chat (LoRA)	0.88	4.84	1.21	1.29	0.82	0.65	0.93	5.85	3.29	4.38	2.41
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	2.13	5.56	2.39	0.52	1.52	0.48	0.40	4.66	1.66	3.24	2.26
	LLaVA-1.5-13B-Chat (LoRA)	0.29	3.93	0.95	0.04	0.40	0.09	0.20	2.55	0.73	1.88	1.11
	LLaVA-NeXT-7B-Chat (LoRA)	0.11	4.68	5.27	0.14	0.74	0.56	0.45	4.14	2.22	3.17	2.15
	LLaVA-NeXT-13B-Chat (LoRA)	0.41	2.17	0.84	0.15	0.08	0.00	0.13	2.15	0.64	2.15	0.87
	Qwen2-VL-7B-Instruct (LoRA)	2.97	0.71	0.12	0.12	0.16	0.46	0.72	3.53	1.75	3.34	1.39
	Qwen2.5-VL-3B-Instruct (LoRA)	0.57	3.86	1.80	0.16	0.70	0.04	0.05	1.93	0.00	2.50	1.16
	Qwen2.5-VL-7B-Instruct (LoRA)	0.75	2.51	1.12	0.09	0.37	0.03	0.19	2.32	1.31	2.85	1.15
	Llama-3.2-11B-Vision-Instruct (LoRA)	0.30	1.92	0.24	0.02	0.46	0.14	0.53	4.79	2.44	7.88	1.87
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	0.01	3.80	3.16	0.26	0.35	0.06	0.14	1.88	0.00	1.61	1.13
	LLaVA-1.5-13B-Chat (MCLF)	0.68	2.54	1.88	0.21	0.16	0.31	0.22	1.74	0.55	1.91	1.02
	LLaVA-NeXT-7B-Chat (MCLF)	1.83	1.74	3.72	0.34	0.21	0.01	0.09	2.71	0.82	2.54	1.40
	LLaVA-NeXT-13B-Chat (MCLF)	0.46	4.64	2.08	0.63	0.11	0.17	0.30	1.65	0.64	1.16	1.18
	Qwen2-VL-7B-Instruct (MCLF)	0.30	3.46	2.02	0.19	0.08	0.25	0.15	0.88	1.35	0.31	0.90
	Qwen2.5-VL-3B-Instruct (MCLF)	0.44	2.53	1.95	0.07	0.42	0.09	0.00	1.76	0.82	1.06	0.91
	Qwen2.5-VL-7B-Instruct (MCLF)	0.77	3.77	1.55	0.26	0.70	0.24	0.15	1.55	0.90	2.28	1.22
	Llama-3.2-11B-Vision-Instruct (MCLF)	0.89	2.76	2.05	0.43	0.14	0.07	0.11	2.69	1.43	3.82	1.44
	Avg. Change of MCLF over LoRA	-0.27	-0.01	+0.71	+0.14	-0.28	-0.07	-0.19	-1.40	-0.53	-1.54	-0.34

Table 1: Performance-normalized hallucination index $\mathcal{H} = \frac{|Prec-Rec|}{Acc+F1}$ ($\times 10^{-2}$, lower is better) of PTMs, LLMs, and MLLMs on metaphor-rich semantic orientation, metaphor-rich pragmatic figurativeness, and partial metaphor reasoning tasks. *Acc* and *F1* are taken from Table 4 of the main paper, and *Prec* and *Rec* (*macro-Prec.* and *macro-Rec.*) are taken from Table 1 of the Supplementary Material. *Avg.* is the mean $\mathcal{H}$ over the ten tasks. Cyan means the second best. Purple means the best. The Random baseline is excluded from the ranking. Green indicates that MCLF reduces the average $\mathcal{H}$ compared with LoRA, while red indicates an increase.